

CORVALLIS 10 SSW, OR, USA

WMO: 726960

Lat: 44.4185N

Lon: 123.3257W

Elev: 95

StdP: 100.19

Time Zone: -8.00 (NAP)

Period: 06-19

WBAN: 04236

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.9	-3.1	-8.8	1.8	-2.2	-5.9	2.3	-1.2	4.6	8.7	4.0	8.1	0.3	N/A	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	32.7	19.5	30.5	19.0	28.5	18.1	20.5	30.4	19.6	29.0	18.7	27.5	1.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	12.1	24.1	15.8	11.4	22.8	15.1	10.8	21.7	59.0	30.3	55.9	29.0	53.1	27.7	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.6	3.0	2.6	-8.2	37.0	3.2	2.0	-10.4	38.4	-12.3	39.6	-14.1	40.7	-16.3	42.1		
(5)			-8.1	22.0	3.4	0.5	-10.5	22.4	-12.5	22.7	-14.4	23.0	-16.9	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	4.5	5.8	7.7	9.6	12.9	15.8	19.3	19.4	16.6	11.5	7.3	4.1
	DBStd	6.18	3.60	3.05	2.80	2.89	3.01	3.11	2.65	2.66	3.03	2.85	3.53	3.83
	HDD10.0	723	173	120	82	42	7	0	0	0	1	16	96	187
	HDD18.3	2726	429	351	330	263	171	88	19	18	69	211	333	442
	CDD10.0	1171	2	3	11	28	96	174	287	293	197	64	13	3
	CDD18.3	132	0	0	0	0	2	13	48	52	17	1	0	0
	CDH23.3	2328	0	0	0	11	79	255	847	808	314	12	0	0
	CDH26.7	853	0	0	0	1	13	81	338	321	98	2	0	0
Wind	WSAvg	1.0	0.9	1.1	1.2	1.1	0.9	0.9	1.0	0.9	0.9	0.9	1.0	1.0
Precipitation	PrecAvg	1168	187	130	132	83	64	36	7	10	32	107	169	210
	PrecMax	2086	433	387	255	214	145	81	30	48	177	279	423	458
	PrecMin	582	39	14	29	24	0	2	0	0	0	26	33	43
	PrecStd	298	104	85	58	46	38	20	8	14	36	54	91	99
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	15.4	19.9	25.2	28.3	32.8	35.7	35.6	32.2	24.8	17.8	14.5
		MCWB	11.7	11.7	13.0	16.7	17.7	20.9	19.1	19.4	18.4	15.6	13.9	12.6
	2%	DB	12.3	13.5	16.7	21.0	25.7	28.9	32.8	32.6	29.6	21.8	15.6	12.8
		MCWB	10.4	10.2	11.5	14.1	16.9	18.9	19.4	19.5	17.9	14.2	12.8	11.4
	5%	DB	11.0	11.9	14.9	17.9	23.0	26.1	30.4	30.3	27.0	19.2	14.0	11.1
		MCWB	9.5	9.5	10.6	12.0	15.6	17.4	19.0	18.9	16.9	13.5	12.0	10.0
	10%	DB	9.9	10.6	13.0	15.5	20.3	23.8	28.1	27.9	24.5	17.0	12.7	9.6
		MCWB	8.6	8.9	9.8	10.9	14.3	16.1	18.1	18.0	16.1	13.2	10.9	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.8	13.6	17.5	19.0	21.7	21.3	21.6	20.4	17.9	15.0	13.0
		MCDB	13.7	14.5	17.5	24.1	26.3	31.2	31.4	32.8	29.1	22.7	16.4	13.7
	2%	WB	11.3	11.5	12.5	14.7	17.4	19.8	20.2	20.3	19.1	16.3	13.6	11.5
		MCDB	12.3	13.0	15.6	20.0	23.9	28.4	30.5	30.6	27.1	19.7	14.9	12.5
	5%	WB	10.0	10.2	11.5	13.0	15.8	18.1	19.1	19.5	17.8	15.0	12.4	10.1
		MCDB	11.0	11.7	14.3	16.9	21.7	25.4	29.2	29.0	25.1	18.2	13.7	11.0
	10%	WB	9.0	9.2	10.4	11.6	14.6	16.7	18.3	18.6	16.7	13.9	11.0	8.6
		MCDB	10.1	10.6	13.0	14.7	19.8	23.3	27.6	27.4	23.1	16.4	12.4	9.5
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.4	9.2	10.5	12.3	13.4	16.3	16.4	14.7	11.0	7.5	5.9
		MCDBR	7.3	9.1	12.5	15.3	17.7	18.0	20.3	20.3	19.5	15.4	8.6	6.6
		MCWBR	5.5	6.0	7.8	8.9	10.0	8.9	8.7	8.4	8.7	8.8	6.2	5.5
	5% WB	MCDBR	6.7	7.2	11.2	13.7	16.1	16.8	19.2	18.8	17.3	11.8	7.4	6.2
		MCWBR	5.4	5.6	7.3	8.7	9.4	8.7	8.8	8.5	8.1	7.0	5.5	5.5
Clear-Sky Solar Irradiance	taub	0.315	0.316	0.322	0.356	0.363	0.356	0.349	0.355	0.344	0.325	0.322	0.326	
	taud	2.542	2.550	2.540	2.405	2.405	2.429	2.467	2.472	2.493	2.557	2.532	2.477	
	Ebn at Noon	814	882	921	907	907	911	914	898	882	857	796	754	
	Edh at Noon	67	78	90	111	115	112	107	103	93	77	67	65	
All-Sky Solar Radiation	RadAvg	1.32	2.13	3.12	4.47	5.69	6.31	7.18	6.15	4.63	2.74	1.49	1.07	
	RadStd	0.19	0.27	0.37	0.37	0.54	0.56	0.42	0.38	0.38	0.31	0.19	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+17

Nomenclature: See separate page